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MODULE TITLE: DATA MINING AND ANALYSTIC

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1.2. What is collinearity? What effect does collinearity amongst predictor features/ variables have on their estimated coefficient value?

Collinearity: Explain what it means when predictor variables are highly correlated with its effect on regression coefficients such as Coefficient stability, Standard errors, Statistical significance, Interpretability

2.1 What is the difference between logistic regression and linear regression?

Difference Between Logistic Regression and Linear Regression

Logistic Regression	Linear Regression
Used when the dependent variable is continuous	Used when the dependent variable is binary
Predicts numeric values (e.g., income, weight, temperature)	Predicts probabilities between 0 and 1
Output can take any real value	Uses the logistic (sigmoid) function
Formula: $y = \beta_0 + \beta_1 x_1 + \cdots + \beta_n x_n$	Formula: $P(y = 1) = \frac{1}{1+e^{-(\beta_0+\beta_1 x_1+\cdots)}}$

3.1 Give a qualitative description of Principal Component Analysis (PCA) and its applications in machine learning. Why might it be useful to consider PCA to transform a set of explanatory variables?

Principal Component Analysis: is a dimensionality reduction technique used to transform a dataset with many correlated variables into a smaller set of uncorrelated variables, called principal components.

Each principal component

- Is a linear combination of the original variables
- Captures the maximum possible variance
- Is orthogonal (independent) from the others

Applications of PCA in Machine Learning

- Dimensionality reduction
- Feature extraction
- Data visualization (2D or 3D plots)
- Noise reduction
- Dealing with multicollinearity
- Speeding up machine learning algorithms

PCA to transform a set of explanatory variables

- Removes redundancy caused by correlated variables
- Reduces the number of features while retaining most information
- Improves model stability and interpretability
- Helps avoid overfitting
- Makes visualization possible in lower dimensions

3.2 Write down the mathematical equations for PCA explaining how one transforms the raw input data matrix X into a new set of variables. Give an interpretation of each matrix.

The mathematical equations for PCA

$X \in \mathbb{R}^{n \times p}$ be the data matrix
where by
"n" observations,
"p" variables

Step 1: Standardization

$$X_{std} = \frac{X - \mu}{\sigma}$$

i. Centers data to zero mean
ii. Scales variables to unit variance

Step 2: Covariance (or Correlation) Matrix

$$\Sigma = \frac{1}{n-1} X_{std}^T X_{std}$$

i. Measures relationships between variables

Step 3: Eigen Decomposition

$$\Sigma v_i = \lambda_i v_i$$

Where:

i. λ_i= eigenvalues (variance explained)
ii. v_i= eigenvectors (principal directions)

Step 4: Projection onto Principal Components

$$Z = X_{std} V$$

i. Z= transformed data (principal components)
ii. V= matrix of eigenvectors